Evaluating AI-based manufacturability decisions against the resolution of production

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Abstract

Design-stage manufacturability decisions rest increasingly on simulation and machine learning, whose fitness to decide is rarely evaluated against production. The proposed framework's unit, the production floor, is the difference that drift and scatter of production and measurement produce between two batches of one design within a run. For each relation of a new design to produced evidence, a rule returning meets, fails or trial has a decisive distance in floors, beyond which at least 95 % of its verdicts are correct, and a safe distance, beyond which at most 5 % are wrong; a procedure adds guard bands for single verdicts. Fourteen rules were evaluated at 95 % conformance on open data of 18 deep-drawn alternatives of 500 parts, mostly process settings, with matched simulations (a draw-in floor: 0.05 mm). With a produced sibling, the nearest produced setting was right for requirements over 3.4 floors from the truth (near-replicates 0.85, resolvable siblings 4.9). For a new setting the best rules needed 4.7 floors interpolating, 4.9 extrapolating to 500 kN and 8.9 to 100 kN, with a stroke-speed change; for a new family the best was safe at 16–17 floors in either family's floor. Requirements at a performance index of 1.33 lay inside every decisive distance except a near-replicate's. With six to nine produced alternatives at two or three forces, the simulation as used, offset or rescaled, helped only across families; learned 90 % intervals covered 85–96 % with a sibling, under 50 % extrapolating or across families. This gives AI-based manufacturability support a production-referenced evaluation standard.

Keywords: machine learning, design for manufacturing, predictive quality, uncertainty quantification, operating characteristic, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming the decision rests on a finite-element simulation (Banabic, 2010): the team compares the predicted characteristics with the requirements and releases the alternative, changes it or sends it to a physical trial. Simulation has moved such decisions forward (Thomke, 1998), and artificial intelligence makes the loop faster: surrogates replace simulations, models learn manufacturability from earlier and synthetic designs, checks are automated (Yan et al, 2026; Ghadai et al, 2018; Zhang et al, 2018; Ballegeer and Benoit, 2026), and predictive-quality models and virtual metrology estimate product quality from process data (Tercan and Meisen, 2022; Dreyfus et al, 2022). Such models are judged by their prediction error on held-out data, which leaves open the question the team has to answer: *can this prediction be relied on to decide?* Mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996), and production variation, with which development organisations still contend (Thornton et al, 2000), adds a further reason: whether a prediction-based decision is credible is a property of production, and production evidence is scarce at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study. How can the fitness of a design-stage prediction to decide be expressed on a common production-referenced scale, for any predictor and characteristic (RQ1)? How does it depend on the amount of production evidence and on how the new design relates to what has been produced (RQ2)? And what do simulation and machine learning contribute once production evidence exists (RQ3)? Uncertainty quantification for machine learning supplies intervals (Nemani et al, 2023), but AI-based decision support, decision-based design (Hazelrigg, 1998), design margins (Eckert et al, 2019) and test planning (Thomke and Bell, 2001; Salado and Kannan, 2018; Tahera et al, 2019) need to know how close to a requirement a prediction can be relied on.

The framework proposed here has the *production floor* as its unit: the difference that drift and scatter of production and measurement produce between two batches of one design within a run, a floor in the sense of a noise floor. A rule that turns the design-stage evidence into a verdict (*meets*, *fails* or a physical *trial*) is characterised, for each *evidence relation* between a new design and the produced alternatives, by two points on its operating characteristics in floors, its *decisive* and *safe distances*, and a procedure turns them into a guard band for single verdicts. The decisions in scope are those of a team that has produced variants of a design family, in process planning and tryout and when production knowledge is carried to a new family.

The evaluation uses open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each (Baum and Heinzelmann, 2026), with finite-element simulations of the same process (Baum and Heinzelmann, 2025). Within each of two cup geometries the alternatives differ in blank-holder force and lubrication on one tool, so two of the three relations concern process settings, and the geometries give two transfers between families. Each alternative is judged as a new design by fourteen rules at a 95 % conformance level, against what its 500 parts did. The study shows whether the framework discriminates between rules and evidence situations, not whether design teams decide better with it, and

its findings on one dataset are hypotheses for confirmation. The contributions are the framework (Section 3), an evaluation design that separates evidence relations and isolates what the simulation adds (Section 4), empirical answers for deep drawing (Section 5), and guidance and a reporting standard for evaluating simulation- and AI-based manufacturability support, including classifiers and language-model checks (Section 6).

2 Background and related work

2.1 Machine learning for manufacturability and quality

Design for manufacturing (DFM) codifies what a process can make into rules, guidelines and cost models (Boothroyd et al, 2010), and quantitative manufacturability models have long driven design decisions (La Trobe-Bateman and Wild, 2003). Machine learning learns manufacturability from CAD models (Ghadai et al, 2018; Zhang et al, 2018), also from synthetic sheet-metal datasets (Ballegeer and Benoit, 2026), from process data and simulated design spaces (Yan et al, 2026), and generative models propose designs directly (Regenwetter et al, 2022). In production, predictive-quality models and virtual metrology estimate quality from process data (Wuest et al, 2016; Tercan and Meisen, 2022; Dreyfus et al, 2022), their reproducibility and reuse have become research questions (Xie et al, 2025, 2026), and for deep drawing, discrepancies between synthetic and real data are being addressed for AI applications (Baum et al, 2026b). Learned models are evaluated by held-out accuracy on labelled designs or parts, and language and vision-language models by case studies (Makatura et al, 2024) and benchmark questions (Picard et al, 2025), not by the decisions they support against the production of the alternatives they are asked about.

2.2 Design decisions, design types and production evidence

Decision-based design casts design as a choice between alternatives with uncertain outcomes (Hazelrigg, 1998). How much a team knows about a new alternative depends on the kind of design, variant, adaptive or original (Pahl et al, 2007), organised across products by platform design (Simpson et al, 2006); the evidence relations of this paper are their counterparts for production evidence, the design type concerning the novelty of the solution and the evidence relation the novelty of its realisation relative to what has been produced. Production knowledge enters design through process-capability data (Thornton, 2004; Tata and Thornton, 1999). Unresolved uncertainty is absorbed by margins (Eckert et al, 2019), whose part that covers uncertainty is distinct from the part that absorbs change (Eckert and Isaksson, 2017). Tests and prototypes, the alternative to a margin, are modelled as a trade between the costs of tests and of errors (Thomke, 1998; Thomke and Bell, 2001; Salado and Kannan, 2018) and studied as design activities (Tahera et al, 2019). When upstream information is fit to act on is the question of overlapped development (Krishnan et al, 1997; Terwiesch et al, 2002), and set-based concurrent engineering codifies production knowledge as feasibility limits (Sobek et al, 1999). All of these assume a measure of how far a design-stage prediction can be relied on; the framework supplies one in a unit set by production.

2.3 Simulation credibility, calibration and validation

Verification and validation ask whether a model is accurate enough for its intended use; the distinction between its validation and application domains (Oberkampf and Roy, 2010) is the question that the evidence relation asks of a design, and risk-informed credibility assessment scales the required evidence with the consequence of a wrong decision (ASME, 2018). Bayesian calibration adds a discrepancy model and calibration parameters to a simulator (Kennedy and O'Hagan, 2001), multi-fidelity models scale a cheap model to an expensive one (Kennedy and O'Hagan, 2000), and a model may be relied on only within its valid input domain (Malak and Paredis, 2010). Conformal prediction gives distribution-free intervals under exchangeability (Lei et al, 2018; Barber et al, 2021), also under covariate shift (Tibshirani et al, 2019) and for decision risks (Angelopoulos et al, 2024), and have entered quality assurance (Heddoub et al, 2026) and engineering design (Zong et al, 2026). In sheet-metal forming, friction (Hol et al, 2012) and the unloading behaviour of the material (Yoshida and Uemori, 2002) are main sources of simulation error, simulation datasets of deep drawing support surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025), and the deviation between simulation and parts has been decomposed statistically (Baum et al, 2026a). How close to a requirement such intervals and accuracy statements can be relied on to decide, and how much production evidence that takes, remains open.

2.4 Decisions with abstention and their operating characteristics

In conformity assessment a measured value proves conformance, proves nonconformance or leaves the decision open, with guard bands set by the measurement uncertainty and the risks of false acceptance and rejection as the measures of a decision rule (ISO, 2017). In selection procedures and acceptance sampling the operating characteristic gives the probability of a decision against the true quality, with an indifference zone inside which no decision is required (Bechhofer, 1954). A classifier with a reject option trades errors against abstentions (Chow, 1970; Geifman and El-Yaniv, 2017), learning to defer passes difficult cases to an expert (Madras et al, 2018), and appropriate reliance on automation requires that its reliability be visible to its users (Lee and See, 2004), also in design teams working with AI (Zhang et al, 2021). The decisive and safe distances are operating characteristics of this kind, and the procedure's guard band is the design-stage counterpart of a conformity-assessment one.

2.5 Production variation and measurement

Statistical quality control compares short- and long-term variation with tolerances through capability and performance indices (Montgomery, 2019), with release criteria such as a performance index of at least 1.33 (Automotive Industry Action Group, 2006) and time-dependent models for drifting processes (ISO, 2026). Precision experiments define the repeatability limit, the difference that two results under repeatability conditions exceed with a probability of 5 % (ISO, 2025), and intermediate precision under conditions that differ in time, operator or equipment (ISO, 2023); measurement

Table 1 Positioning among methods that support or evaluate design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Design margins (Eckert et al, 2019)	experience, analysis	buffer to the requirement	rarely	no
V&V, credibility assessment (Oberkampf and Roy, 2010; ASME, 2018)	physics model, validation tests	accuracy in physical units; required evidence	on test cases	can require tests
Bayesian calibration (Kennedy and O'Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage; per decision possible	yes
Conformity assessment, OC curves (ISO, 2017; Bechhofer, 1954)	measurement and its uncertainty	accept, reject, undecided; risks	on the measured item	yes
Selective classification (Chow, 1970; Geifman and El-Yaniv, 2017)	labelled data	class or reject; risk–coverage	on held-out data	yes
ML manufacturability models (Ghadai et al, 2018; Yan et al, 2026)	historical or simulated designs	class or value	on held-out data	rarely
This paper	simulations, produced alternatives	verdict; decisive and safe distance in floors by evidence relation	yes, per verdict	yes

systems analysis separates repeatability from stability (Automotive Industry Action Group, 2010). The production floor is a precision limit of this kind for batch centres within a run (Section 3.2); what is new is its use not to judge whether two results agree but as the unit in which a design decision rule is scored and its guard band set. Table 1 places the framework among them: it does not replace calibration or conformal prediction but scores them as decision rules against production in a common unit.

3 The framework

3.1 The decision problem and its scope

A design alternative a is a combination of product geometry and process settings produced as a series of parts, over which a quality characteristic c varies. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet: alternative a is *adequate* for c when the p-quantile of its parts satisfies $q_p(a, c) \leq R$. Lower or two-sided limits follow by a change of sign or by the absolute deviation from the target. At the design stage q_p is unknown. A decision rule r maps the available evidence to an interval

$[L_r, U_r]$ for q_p and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \le R, \\ \text{fails} & L_r > R, \\ \text{trial} & \text{otherwise.} \end{cases} \tag{1}$$

The three verdicts mirror the decision rules of conformity assessment (ISO, 2017), here for the uncertainty of a design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*. The demonstration decides on $p = 0.95$, which for a normal characteristic corresponds to a performance index of about 0.55. Release criteria of series production, a performance index of 1.33 or more (Automotive Industry Action Group, 2006), concern nonconforming fractions in parts per million that a series of 500 parts cannot estimate without a parametric tail; Section 5.8 locates such limits relative to the distances, without deciding them.

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts, each cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10 % trimmed mean). The production floor of characteristic c is

$$F_c = Q_\alpha\big\{\, |m_i - m_j| : i < j \text{ batches of the same alternative} \,\big\}, \tag{2}$$

the α-quantile of the difference between two batches of one design. Under a stationary normal model with a standard deviation σ_b between batch means and σ_w between parts, $F_c = 1.96\sqrt{2}\sqrt{\sigma_\mathrm{b}^2 + \sigma_\mathrm{w}^2/B}$ at $\alpha = 0.95$, the form of the repeatability limit of precision experiments (ISO, 2025). Because it pools batch pairs up to 450 parts apart and grows with their lag (Section 5.1), it is closer to a time-different intermediate-precision limit (ISO, 2023) of batch centres within a run; in the terms of ISO 22514-2 the runs follow a time-dependent model with varying location (ISO, 2026). The reference protocol of this paper is $B = 50$, ten batches of a 500-part run, all pairs of batches and $\alpha = 0.95$, with per-geometry floors where geometries differ. The floor rests on three assumptions: batches of one design differ only by drift and scatter within a run (A1); the variation within the runs from which it is estimated is like that within future runs (A2); and the measuring system is repeatable and its measurement resolution small against the floor (A3, checked in Section 5.1).

Four properties matter for its use. It has the unit of the characteristic, so distances in floors are on a common scale for all characteristics and predictors, though equal distances do not mean equal nonconforming fractions (Table 3). It is a property of a protocol: under drift it grows with the time between the batches compared and shrinks with B, and σ_b and σ_w are reported so that it can be recomputed. It contains whatever drifts between batches, process and measuring system alike, so without a reference artefact scanned through the series it is a production-and-measurement floor. And it

excludes the variation between runs, which makes it a lower bound of the variation of series production. A new family, whose floor is unknown before it is produced, is scored in the floor of the produced family, and because the floor resolves batch centres, the reproducibility of the decided quantile is reported alongside.

Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\mathrm{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| \,/\, F_c \tag{3}$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative; this is a practical threshold, not a test. For a continuous design or process parameter x with a local sensitivity $s_c = \partial\bar{m}/\partial x$, the minimum resolvable change is $F_c/|s_c|$: a change of x by less moves the centre of the characteristic by less than two batches of one design differ. It is local and concerns centres; a change below it can still move the tail.

3.3 Decision rules

Fourteen rules span what a design team can do with its evidence (Table 2); the first seven and the force trend M1sn carry the main results. Beside the nominal simulation s_0 (M0), the bias-corrected simulation (M1) adds the median offset between the calibration alternatives' q_p and their s_0; the scaled simulation (M1s) fits $q_p = a + b\,s_0$ to them by least squares, a linear multi-fidelity correction (Kennedy and O'Hagan, 2000). The envelope rule (M2) adds the median offset and the largest absolute residual of the calibration alternatives to the upper end of the simulation envelope over plausible incoming conditions, which vary sheet thickness and friction but not material properties. M5 models the offset between q_p and the envelope as a Gaussian process (Rasmussen and Williams, 2006) with a 90 % predictive interval, whose noise variance is bounded below by the variance between the calibration alternatives' series of one geometry and force, so that a new series of a produced setting is not predicted more precisely than produced series reproduce each other; M5n is the same process of q_p without the simulation. The nearest produced setting (NN) is a capability-database look-up that uses neither simulation nor learning. The seven further rules include a friction-calibrated simulation (Kennedy and O'Hagan, 2001) (Mc), part-level gradient boosting (Friedman, 2001) with a jackknife+ interval (Barber et al, 2021), with and without the simulation (M3, M3n), and the least-squares line of q_p in force (M1sn). The nominal simulation and the envelope do not depend on the lubrication pattern, so within a geometry they are functions of the force alone: M1sn is the counterpart of M1s without the simulation, and a lubrication variant gives M0, M1, M1s, M2 and M5 no simulated information that distinguishes it from its siblings.

With n calibration alternatives, the largest-residual margin covers a new exchangeable residual with probability $n/(n+1)$ for a fixed offset and at least $(n-1)/(n+1)$ for an estimated one; it is the split-conformal quantile (Lei et al, 2018) at the highest attainable level, not a 90 % interval. A distribution-free interval cannot certify a miss rate below $1/(n+1)$, so a miss rate of 5 % needs at least 19 produced alternatives in the relevant relation. With six to nine calibration alternatives the jackknife+ bounds at the level of 0.9 fall at or beyond the extreme leave-one-out values, which are used instead; under exchangeability these guarantee a coverage of $1-2/(n+1)$, 0.71 to 0.80

Table 2 The decision rules; the first seven and M1sn carry the main results

Rule	Prediction of q_p	Interval	Evidence
M0 nominal simulation	simulation at nominal thickness and friction	point	none
M1 bias-corrected simulation	M0 plus the median offset	point	q_p
M1s scaled simulation	$a + b$ M0, least squares	point	q_p
M2 envelope + residual margin	envelope top plus the median offset	largest residual	q_p
M5 GP-corrected envelope	envelope plus a GP offset over the descriptors	GP, 90 %	q_p
M5n GP of q_p	GP of q_p over the descriptors	GP, 90 %	q_p
NN nearest produced setting	mean q_p at the nearest produced setting	point	q_p
M0w process-window worst case	simulated range over the incoming conditions	range	none
Mc friction-calibrated simulation	friction calibrated on the draw-in, plus offset	point	centres, q_p
M3 simulation + boosting	matched simulation plus boosted discrepancy	jackknife+	parts
M3n boosting, no simulation	boosted model of the characteristic	jackknife+	parts
M1n median of the produced	median q_p of the calibration alternatives	point	q_p
M1sn force trend	$a + b$ force, least squares	point	q_p
M2n median + residual margin	M1n with the largest-residual margin	largest residual	q_p

Fitted on the calibration alternatives. Descriptors: geometry indicator, blank-holder force and lubrication rank (the order of the oil amount). GP: Gaussian process. M3 and M3n at part level; M3n is named M4 in the released code. M6, a conformalised GP, is a robustness variant (Table 4).

(Barber et al, 2021), approximately because the residuals are centred on their median. The Gaussian-process interval is model-based. The level trades decisiveness for safety, as the cost of errors would set the threshold of a reject option (Chow, 1970); 0.9 is used here, 0.8 and 0.95 in the robustness analysis.

3.4 Operating characteristics: decisive and safe distances

A rule is scored by judging each produced alternative as a new design at requirements placed at a known distance from the truth,

$$R = q_p(a, c) + d\,F_c, \tag{4}$$

so that a is adequate exactly when $d \geq 0$. For a case (a, c) with interval $[L, U]$, let $u = (U - q_p)/F_c$ and $v = (q_p - L)/F_c$ be the distances by which the interval excludes the truth from above and below. At $d = +\delta$ the verdict is correct when $u \leq \delta$ and a false reject when $-v > \delta$; at $d = -\delta$ it is correct when $v < \delta$ and a false accept when $-u \geq \delta$. Let $\kappa(\delta)$ and $\omega(\delta)$ be the shares of correct and of wrong verdicts among all feasible verdicts at $|d| = \delta$: every case gives one verdict on an adequate alternative and, where $q_p - \delta F_c \geq 0$, one on a failing alternative, and every verdict has equal

weight. As δ grows, failing-side requirements become infeasible for characteristics close to zero, so the failing side carries half the weight up to a few floors and a third at about 100 floors (98 of 126 failing-side requirements remain feasible at 16 floors and 63 at 108). The *decisive distance* δ_{d} of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_{s} the smallest δ beyond which $\omega \leq 0.05$.

The distances are exact up to the 0.05-floor grid on which they are evaluated: κ and ω change only where δ crosses one of the case's u, v, $-u$ or $-v$, so the distances are order statistics of these exclusion distances; for a rule that always decides, $\delta_{\mathrm{d}} = \delta_{\mathrm{s}}$ is close to the 90 % quantile of its absolute error in floors. Always $\delta_{\mathrm{s}} \leq \delta_{\mathrm{d}}$, and between the two the rule sends designs to trial rather than misleading; one-sided distances for failing alternatives (false accepts) and adequate ones (false rejects) matter when the errors cost differently. The distances are points on operating characteristics over a population of design situations: δ_{d} is the analogue of the indifference zone of a selection procedure (Bechhofer, 1954) at a probability of correct selection of 0.95. A trial is not error-free: a short run reproduces the 95th percentile only to about one floor within a run (Section 5.1) and, being a run, also carries the variation between runs, so no decision is reliable closer to the requirement than the trial replacing it.

3.5 Evidence relation, calibration budget and applicability guard

How a new design relates to the produced alternatives is part of the decision situation. With a *sibling*, an alternative at the same process setting has been produced that differs only in the factor varied at a fixed setting (here the lubrication pattern), which production may not resolve; a *new process setting* lies between produced settings (interpolation) or outside them (extrapolation); a *new family* shares no produced alternative. The calibration budget is the dependence of the distances on the number k of produced alternatives, over all subsets of size k classified by relation. An applicability guard flags a new design that lies outside the calibration: its geometry is not calibrated (family guard), a descriptor lies outside the calibrated range (range guard), or its simulated q_p lies outside the calibration alternatives' simulated values (novelty guard), in the spirit of a valid input domain (Malak and Paredis, 2010).

3.6 Procedure

Figure 1 turns the framework into six steps: steps 1 to 5 once per design family, repeated when production evidence is added, and step 6 for every new design. Steps 1 to 4 frame the decision (family, alternatives, characteristics, form of the requirement, conformance level), measure the floor with a stated protocol, state which differences production resolves and calibrate the candidate rules with and without the simulation. Step 5 qualifies them by their distances for each evidence relation and fixes for each rule and relation a *guard band* g: the smallest observable margin between the predicted interval and the requirement beyond which at least 95 % of the decided verdicts were correct. Being conditional on what is observed, g depends on the distribution of requirement distances that the team expects, not on rule and relation alone. The reference prior weights the 47 distances of the requirement grid within ± 20 floors equally

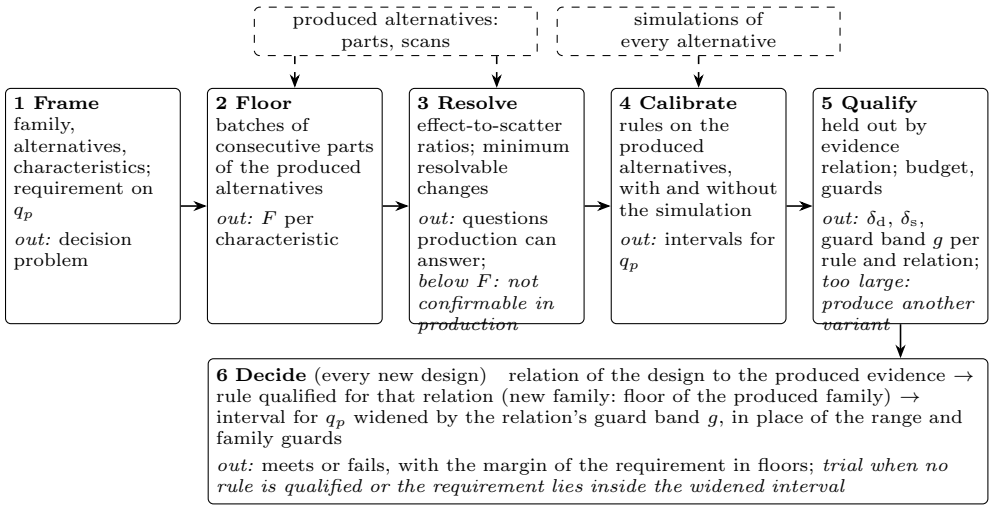


Fig. 1 The procedure: steps 1–5 once per design family, step 6 for every new design; dashed boxes are the evidence it uses

(0, ±0.5 to ±10 in steps of 0.5, ±12, ±15 and ±20; 87 % of the weight within ±10), drops infeasible requirements and searches g in steps of 0.25 floors up to 20, beyond which the rules send most designs within the prior's range to trial; it is a reference for requirements set from a few to some tens of floors from the true 95th percentile, denser where verdicts are hardest, which a team replaces by the distribution of its own requirements. Step 6 establishes the relation of the new design, applies the rule qualified for it, widens the rule's interval by g floors and returns meets or fails with the margin in floors, or a trial when no rule is qualified or the requirement lies inside the widened interval. The guard band of the relation takes the place of the range and family guards, which here coincide with the extrapolated and new-family relations; the novelty guard is reported as a diagnostic, not applied, since most of its refusals would have been right; Section 5.7 reports g under other priors and works one decision through.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). Square blanks of DP600 steel, 210 mm wide and about 1 mm thick, were deep drawn into cups (OP10) and cut (OP20) on one tool set with two geometries, concave and convex side walls, which differ in base curvature, wall angle and bottom radius at once. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine, with series medians of 1.15–1.48, 0.92–1.23 and 0.85–1.13 g/m^2 of oil): 18 alternatives, each produced as one series of 500 consecutive parts, the change of geometry being the only change of product design. Every

part has press signals, sheet and oil-film traverses and 3D scans after both operations. The series warm up by 1.0–5.0 K over their first 150 parts from punch temperatures of 20.5–29.6 °C, their median sheet thickness is 0.986–0.993 mm, and the 100 kN series ran at 47 mm/s against 78–79 mm/s. The documentation states neither the order and dates of the series nor whether the parts were scanned in production order.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025): LS-DYNA quarter models with nodes about 1 mm apart, a scaled DP600 flow curve, constant Coulomb friction and springback after both operations; the yield criterion and unloading model, which govern springback (Yoshida and Uemori, 2002), are not documented, and the stroke speed is not represented. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm) and the friction coefficient (0.05–0.15) at each geometry and force. A part is matched to the simulation with the nearest thickness and the friction coefficient that the dataset's linear mapping assigns to its oil film, an assumption of the dataset's authors on which Mc, M3 and every matched simulation depend. The nominal simulation of an alternative is at 0.99 mm, the grid value nearest the median thickness, and friction 0.10.

4.3 Quality characteristics and their measurement

Seven characteristics were extracted with the same definitions from the scans and the simulated nodes, referred to the plane of the cup bottom: the mid-side and corner draw-in and the flange waviness, process indicators on the flange that the cut removes, and four characteristics a drawing would carry: the wall angle (design angle 20° concave, 10° convex), the cup depth, and the arm angle and bottom dome of the cut part, the last two with requirements on absolute values. The four sides of a part measure several of them twice, which exposes the measurement: the flange width along the scan lines scatters 4 to 18 times more than across them, so the mid-side draw-in is taken across the scan lines, and the wall angle is the mean of the east and west walls. The mid-side draw-in is quantised in steps of 0.019 mm (0.38 floors) by the pixel pitch. Differences between scans and simulations, such as the scanned surface against the shell mid-surface, enter the offsets between the two but not the floors.

4.4 Evaluation design

Every produced alternative is judged as a new design under grouped cross-validation (Roberts et al, 2017), one scheme per evidence relation. *New variant*: the alternative alone is held out and the other eight of its geometry calibrate, two siblings at its force among them. *New process setting*: the three alternatives of one force of a geometry are held out and the six others calibrate; 300 kN is interpolated, 100 and 500 kN extrapolated. *New family*: one geometry calibrates the rules for the other (two transfers). Pooled versions of the first two add the other geometry's alternatives to the calibration set. Every calibrated rule is evaluated with and without the simulation (M1/M1n, M1s/M1sn, M2/M2n, M3/M3n, M5/M5n), isolating what the simulation contributes,

and the nearest produced setting is the baseline without simulation and learning. A scope comprises 126 cases (18 alternatives, 7 characteristics).

The intervals of the distances come from 1000 resamples of the held-out alternatives within each geometry, each with a block-bootstrap replicate of every true q_p, common to all rules so that differences get paired intervals; they hold the fitted rules fixed. A second bootstrap refits every alternative-level rule after drawing the three lubrication patterns of each geometry with replacement, at least two of them distinct, which keeps every case in its evidence relation (Table A2). Its 200 resamples contain 48 of the 49 distinct configurations; in three of four draws a pattern is duplicated, so that a new variant keeps one distinct sibling and five distinct calibration alternatives, and the duplicated series enter the noise floor of the Gaussian processes as identical pairs. Its intervals thus show the effect of losing a pattern, not sampling uncertainty.

4.5 Computation and development record

The constants of the floor and the requirement grid were fixed before the decision results were seen, as the repository history records. The rules were not: M0–M5 were defined first; M0w, Mc, M1n, NN, M2n, M5n, the grouped schemes, the exact estimator and the noise floor of the Gaussian processes were added after the first results; an internal review of that version, simulated with a large language model, added M1s, the jackknife+ intervals, the between-run noise floor and the analyses of Sections 5.4 and 5.7; and a second such review added M1sn, the guard bands under other priors and the new family in the produced family's floor. The comparison of rules is therefore exploratory and carries the optimism of a selection made on the evaluation data (Cawley and Talbot, 2010); every variant tried is reported in the paper or in the archived results (Canseven, 2026), which Table A5 maps to the text. A large language model (Claude, Anthropic) assisted under the author's direction with code development, literature retrieval, data analysis, drafting and these reviews; the author designed the study and verified every analysis and reference.

5 Results

5.1 What the production floor contains

In 95 % of batch pairs, two batches of 50 consecutive parts of one alternative differ by up to 0.051 mm in mid-side draw-in and 0.078° in wall angle (Table 3); one floor is 0.7 to 1.7 long-term standard deviations of the parts. The floors are set by drift within the run: without the time order they shrink by a factor of 2.1 to 4.5, and batches one position apart differ by 0.51–0.80 floors and nine apart by 1.1–1.5 (averages over the two geometries). The normal model of Section 3.2 reproduces the floors to within 12 %; for subgroups of five parts, as sampled inspection would measure, it gives floors 1.1 to 2.1 times as large (median 1.3). Independent part-to-part noise, scanner repeatability included, would produce 15–33 % of each floor, and the measurement resolution is small against the floor except for the mid-side draw-in; A3 holds for repeatability and measurement resolution, but the stability of the scanner cannot be checked without a reference artefact. Nor do the process signals identify the source of the drift: batch

Table 3 Production floor per characteristic, pooled and per geometry, with its components

Characteristic	F	95 % interval	Concave	Convex	F/σ_{LT}	σ_{b}/F	σ_{w}/F	Drift
Draw-in, mid-side (mm)	0.051	0.045–0.061	0.050	0.052	1.0/1.4	0.32/0.34	0.91/0.66	3.3
Draw-in, corner (mm)	0.046	0.035–0.052	0.053	0.030	1.4/1.2	0.31/0.35	0.65/0.77	4.5
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0037	0.0021	1.6/0.7	0.33/0.30	0.55/1.31	2.5
Wall angle (°)	0.078	0.066–0.10	0.081	0.078	1.3/0.7	0.32/0.26	0.68/1.39	2.1
Arm angle, cut (°)	0.11	0.094–0.12	0.12	0.100	1.7/0.8	0.36/0.31	0.46/1.22	2.6
Cup depth (mm)	0.011	0.0096–0.012	0.0087	0.012	1.4/1.6	0.31/0.34	0.65/0.52	3.9
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0034	0.0082	1.2/1.6	0.33/0.36	0.76/0.55	3.9

Batches of 50 consecutive parts, 95 % quantile of the difference of their 10 % trimmed means; interval from 2000 batch resamples. Ratios per geometry (concave/convex), classical statistics pooled over the series as root mean squares: σ_{LT}, standard deviation of the parts of a series; σ_{b}, between batch means, corrected for $\sigma_{\mathrm{w}}^2/50$; σ_{w}, within batches. Drift: floor over that of randomly permuted series.

centres of press force, punch temperature, sheet thickness and oil film predict it out of fold with R^2 of -0.23 to 0.20. Variation between runs is not in the floor: the three lubrication series of one geometry and force differ by up to 1.2 (waviness) to 9.8 floors (dome), and a floor taken across them is 1.5 to 9.1 times the reference. Within a run the decided statistic reproduces to about one floor (the 95th percentiles of two batches of 100 parts differ by 0.92–1.49 floors in 95 % of pairs), so distances below about one floor are beyond what production can confirm, and a trial run, which also carries the variation between runs, confirms less still.

5.2 What production resolves and how far the simulation misses it

All 63 geometry contrasts exceed the floor and blank-holder force is resolved in 105 of 126 contrasts, but lubrication only in 51 of 126, so a lubrication variant is often a near-replicate of its siblings. The minimum resolvable force change is 21–67 kN between 100 and 300 kN for most characteristics and lies beyond either force interval for the wall angle and the dome. The nominal simulation misses production by far more than a floor: its median offset reaches 109 floors, its force effect on the draw-in is 2.7 to 5.1 times the measured one, and the cut arms of the concave cups bend downwards in 89–100 % of the simulations but upwards in every part. Part of the offset is definitional: the cup depth differs by a median 0.90 and 0.95 mm, consistent with the scanned surface against the shell mid-surface, and removing this offset alone brings the decisive distance of the nominal simulation from 108 to 65 floors.

5.3 Decisions by evidence relation

Table 4 and Fig. 2 score the rules. The nominal simulation decides at 108 floors (101–115), erring mostly by rejecting adequate alternatives.

A new variant with a produced sibling. Every calibrated rule improves on it, and the simplest is the most decisive: the nearest produced setting, the mean q_{95} of the two siblings, is right beyond 3.4 floors (2.0–4.5), 0.17 mm of mid-side draw-in; the Gaussian processes decide at 7.2 floors but are safe below one floor.

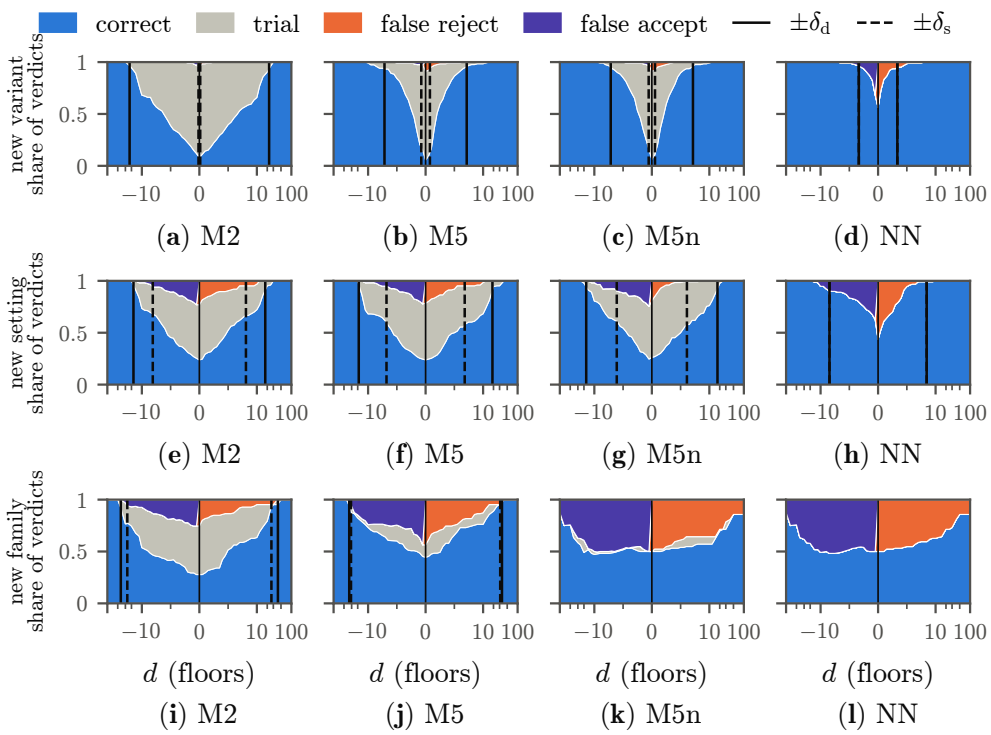


Fig. 2 Verdict shares against the requirement distance d (symmetric logarithmic axis, linear within ±10 floors; 126 cases per panel; a new family in the produced family's floor), with δ_d and δ_s

A new process setting. With every alternative at the new design's force held out, all rules lose: the force trend and the nearest produced setting decide at 8.2 and 8.5 floors, the interval rules (M2, M2n, M3, M3n, M5, M5n) only at 13–31, though they are safe at 5.9–16. Interpolating in force is much easier than extrapolating (the nearest setting 4.7 against 9.5 floors), and the scaled simulation, extrapolating a fitted trend, falsely accepts failing alternatives as far as 164 floors from their limit.

A new design family. Scored in the floor of the produced family, as the procedure prescribes, every rule without the simulation fails in the two transfers (130–135 floors): nothing in one family's production record describes the other. With the simulation, the bias correction decides at 33 floors and M5 at 35, and M3 is safe at 17 floors, M2 at 26 and M5 at 31; in the floor of the new family, which a team would not have, M2 is safe at 16, M3 at 21 and M5 at 28. The best rule was thus safe at about 16–17 floors in either floor, some 0.8 mm of mid-side draw-in, but two transfers do not resolve which rule is safest: the safe distance of M3 minus that of M2 lies between −16 and +7 floors in the produced family's floor and between −3 and +8 in the new family's.

5.4 Where the distances come from

Three features of the design shape these distances. First, the sibling relation mixes replication with variation: in 58 of the 126 cases neither sibling differs resolvably from

Table 4 Decisive and safe distances (floors) by the relation of the new design to the produced evidence

	New variant		New process setting				New family		Pooled	
Rule	δ_d/δ_s	FA	δ_d/δ_s	FA	interp.	extrap.	δ_d/δ_s	FA	variant	setting
M0	108	17	108	17	109	106	104	28	108	108
M1	15	15	24	24	11	25	33	35	27	28
interval	12–18		21–25		10–11	24–25	31–36			
M1s	4.9	4.7	19	164	19	19	121	125	15	20
interval	4.7–6.3		14–21		19–20	15–19	119–121			
M2	23/0.15	0.30	18/8.1	8.3	18/2.7	18/9.9	41/26	27	53/0	53/1.0
interval	22–23		17–18		15–18	17–19	39–42/9.8–32			
M5	7.2/0.75	0.15	19/6.8	7.6	19/2.8	19/8.9	35/31	33	11/0.05	28/5.4
interval	6.6–8.4		18–20		18–20	18–20	31–39/28–32			
M5n	7.2/0.55	0.10	17/6.1	9.0	16/0	18/8.6	133/128	127	9.8/0.20	16/2.7
interval	6.4–7.5		16–18		15–16	16–19	131–133/127–128			
NN	3.4	2.8	8.5	10	4.7	9.5	130	129	3.4	8.5
interval	2.0–4.5		5.5–10		4.3–5.1	6.6–11	129–131			
M0w	119/94	9.2	119/94	9.2	119/94	119/77	117/79	10	119/94	119/94
Mc	27	28	39	39	14	40	43	53	32	33
M3	23/0	0	31/16	13	29/13	33/17	40/17	18	25/0	32/5.1
interval	17–26		25–36		24–31	25–40	36–45/15–19			
M3n	11/0.05	0	13/7.4	9.2	15/7.0	13/7.8	135/126	124	15/0	15/4.9
M1n	9.1	10	10	12	4.8	12	131	129	130	130
M1sn	3.5	3.5	8.2	9.0	4.7	8.9	131	129	77	111
interval	3.1–4.2		6.7–9.0		4.3–5.1	8.6–9.5	129–131			
M2n	19/0	0.20	15/5.9	9.2	13/0	16/9.1	133/128	126	325/0	325/4.0
M6	17/0.05	0	56/6.2	5.8	47/0	56/8.1	35/28	28	–	–

Over all seven characteristics; one number for rules that always decide. FA: safe distance on the false-accept side (failing alternatives). Interval: bootstrap 95 % interval of the decisive distance and, for a new family, of the safe distance (fits fixed). 126 cases per relation (interp.: 42 interpolated; extrap.: 84 extrapolated); a new family rests on two transfers and is scored in the floor of the produced family (Section 3.2). Pooled: the other geometry's alternatives added to the calibration set. Below the line, the further rules (M1sn: the force trend, the counterpart of M1s without the simulation) and, as a robustness variant, M6 (the M5 mean with a conformal margin on standardised leave-one-out residuals).

the held-out alternative, and the nearest setting is then right beyond 0.85 floors, the reproducibility of q_{95} between near-replicate runs; it needs 2.5 floors when one sibling differs resolvably (34 cases) and 4.9 when both do (34 cases), where the force trend needs 4.5 and M5 9.7. Second, the extrapolation is confounded with stroke speed: the nearest setting decides at 11 floors at 100 kN, where the press also ran slower, against 4.9 at 500 kN (the force trend at 8.9 at both). Taken together, the nearest setting decided at up to about 5 floors whenever the new design differed resolvably and force and speed did not change together, though each stratum has 34 to 58 cases and each distance rests on a handful of verdicts. Third, the other family's record does not help with a produced sibling: pooling it makes most rules less decisive (M5 from 7.2 to 11 floors), though it raises M5's coverage from 86 to 90 %.

The relation therefore does not govern decision fitness in general. Over the eleven calibrated rules and three relations, the relation explains 67 % of the variation of the log decisive distance and the rule 6 % (69 and 6 % in the new family's own floor), but

only because rules without the simulation fail for a new family by construction; over the two relations within a family, the rule explains 68 % and the relation 15 % (71 and 15 % with M1sn). Pooled over strata, the best attainable distance grows with novelty: 3.4 floors with a produced sibling, 8.2 for a new setting, 33 for a new family.

5.5 What the simulation and learning contribute

Pairing each calibrated rule with its counterpart without the simulation isolates what the simulation adds as used. Added as an offset, it adds nothing with a produced sibling and degrades two rules: M5 decides where M5n does (difference 0.0 floors, −0.4 to +1.1), while M2 and M1 decide 3.9 (+0.5 to +7.4) and 6.2 floors (+4.6 to +9.7) further out than M2n and M1n, and M1 13.5 further out for a new setting, because the simulated force effect is too large. Rescaled, it does no better: within a geometry the nominal simulation is a function of the force, and the straight line in force (M1sn) decides 1.4 floors closer than the scaled simulation with a sibling (+0.65 to +3.0) and 11 closer for a new setting (+5.3 to +13.8), in every resample and refit configuration; the 4.2 floors that the scaled simulation gains over the median of the produced alternatives with a sibling come from the force level. Only for a new family is the simulation decisive, by 92 to 98 floors as an offset and 10 rescaled, and even there production knowledge carries over where a drawing nominal transfers: for the wall angle, the design angle plus the produced family's deviation from its own decides at 1.9 floors, against 7.5 for M5 (for the arm angle and the dome, whose nominal is zero in both families, this baseline is M1n, at 44 and 27 floors). A Gaussian-process surrogate of the simulations over force, friction and thickness, with a leave-one-out root-mean-square error below 2 floors for 12 of the 14 characteristics and geometries, decides as the simulation does (109 against 108 floors as M0, 15 against 15 as M1 with a sibling).

Learning adds intervals rather than accuracy. Used as point rules, the centres of the M5, M5n and M3n intervals decide at 3.3, 3.1 and 3.7 floors with a produced sibling, as close as the nearest setting (Table A3). With a sibling the Gaussian-process intervals cover 85–86 % of the true q_{95} against a nominal 90 %, but their noise variance sits at its lower bound, the variance between the calibration series of one setting, in 97–99 % of these fits and in every new-setting and new-family fit, so this coverage comes from the quasi-replicate series rather than from the learner; the largest-residual margin covers 89 % and the jackknife+ intervals 96 % (M3) and 89 % (M3n), above their guarantee of 7/9. For a new setting the calibrated intervals cover 43–55 % (M2n and M5n 93 % interpolating), for a new family the Gaussian processes at most 12 %: trees do not extrapolate, two force levels identify at most a linear trend, and in the median new-setting fit the length scale of the force sits at its lower bound.

5.6 Calibration budget

A single produced sibling decides (Fig. 3): with one alternative at the new design's force, the bias correction, the envelope rule and the nearest setting coincide and decide at 3.5 floors (2.3–4.6) without a wrong verdict at 10 floors. Further alternatives do not improve the nearest setting; they make the bias correction worse (12–18 floors), the

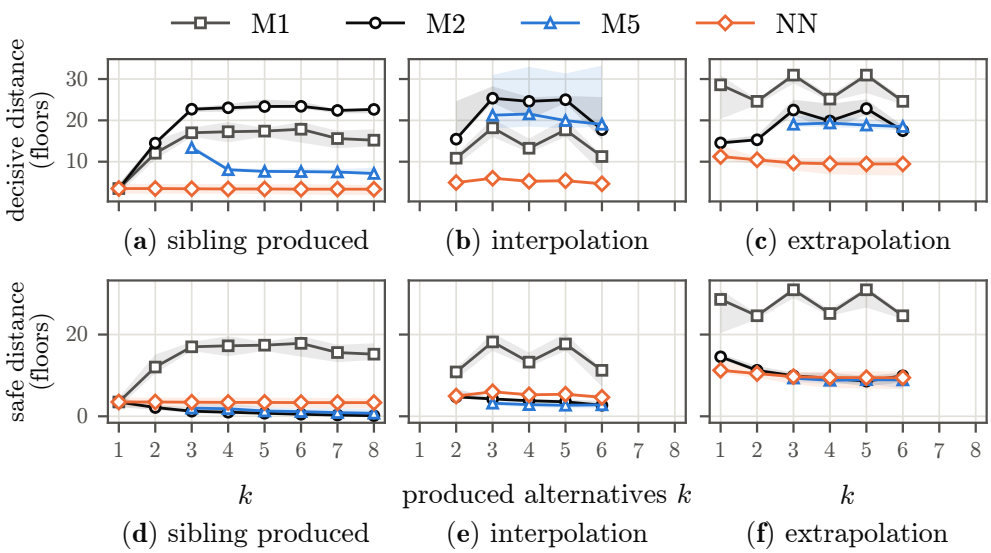


Fig. 3 Decisive and safe distances against the number of produced alternatives of the family, by the relation of the new design to them; bands: 95 % intervals over the held-out alternatives

envelope rule safer but less decisive, and M5, shown from three alternatives because a Gaussian process fits its hyperparameters to them, safer and more decisive (13 to 7.2 floors). Without a sibling, bracketing the new force buys safety but not decisiveness: the envelope rule and M5 are safe at 2.7–4.8 floors but decide only at 15–25, against 4.7–6.0 for the nearest setting, and extrapolation stays unsafe at every budget. With calibration series of 50, 100 or 250 parts instead of 500, no distance within a family moves by more than 1.5 floors, and for a new family by at most 4 in its own floor (Table A2): rules can be calibrated on short runs, but the floors and the held-out truths came from the full series, and whether truths from short runs suffice to qualify the rules was not tested.

5.7 The procedure's decision rule

What a designer observes for a single verdict is the margin between the predicted interval and the requirement, and its value as evidence depends on the relation (Fig. 4): with a produced sibling at least 95 % of the decided verdicts of the interval rules are correct at every margin, for a new setting the share grows with the margin, and for a new family M2 and M5 are right in at most 79 % of their decided verdicts.

Step 5 turns this into guard bands (Table A4). Under the reference prior the interval rules need none with a sibling and the nearest setting 0.25 floors; for an interpolated setting the nearest setting needs 2.0 floors and then decides at 6.7 floors without trials at 10 floors, whereas M2 and M5 need 0.75 and 0.5 floors but send 31 and 40 % to trial; for an extrapolated setting it needs 7.0 floors and sends a quarter to trial. M0 and M0w never qualify, nor do M1 and Mc with a sibling, M1s for an interpolated setting, M1, M1s, M2, M5 and Mc for an extrapolated one, or any rule for a new family.

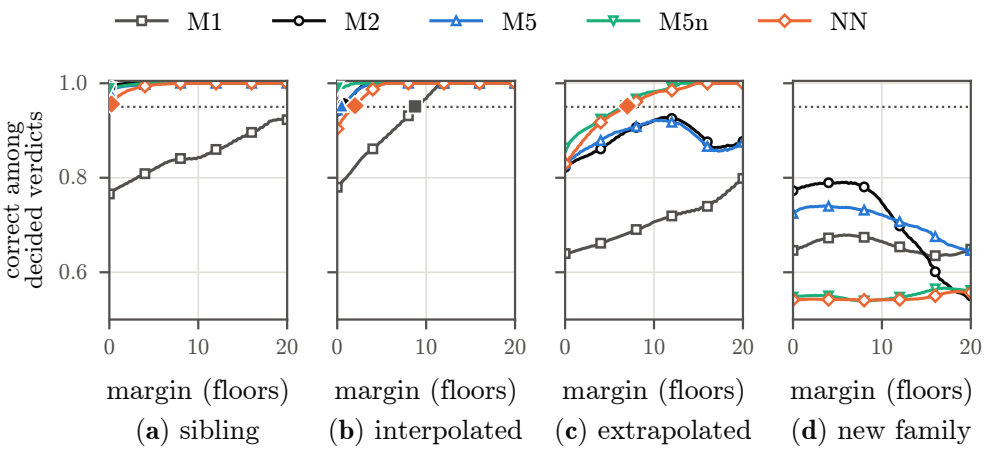


Fig. 4 Share correct among decided verdicts with at least the given margin; reference prior: 47 grid distances within ±20 floors, equal weights; new family in the produced family's floor; filled markers: g

These bands belong to the prior as much as to the rules. A prior concentrated near the truth, where release-level requirements lie (the grid within ±5 floors), raises the band of the nearest setting to 2.0 floors with a sibling, 4.25 interpolating and 14.25 extrapolating (8.0, 5.0 and 14.75 within ±3); a uniform prior within ±20 floors lowers them to 0, 0.25 and 3.5 floors and qualifies M2 and M5 for an extrapolated setting (6.0 and 5.0 floors), and one within ±50 floors qualifies M2, M3 and M5 for a new family (6.25, 6.5 and 15 floors). That every new-family design goes to trial thus follows from the reference prior and the 20-floor cap, as the practice of trying out every new die would have it. The bands are chosen on the evaluation data, which makes the trial rates optimistic. The guards, reported as diagnostics, refuse indiscriminately: the range guard refuses exactly the extrapolated designs (84 of the 126 new-setting cases), and at requirements 10 floors from the truth 96 % of its refusals would have been right for the nearest setting and 72 % for M5 (novelty guard 96 and 73 %).

A worked example. A tryout engineer has to decide whether the concave cup at 300 kN with medium lubrication, a new setting, can be released without a trial run, with the six alternatives at 100 and 500 kN produced and a requirement, chosen for illustration, that 95 % of the parts have a mid-side draw-in, by which tryout sets the blank-holder force, of at most 15.30 mm. The floor is 0.050 mm, the relation is interpolation and no guard fires. Alone, the nominal simulation predicts 15.66 mm and fails the design. The nearest produced setting, the mean of the six alternatives at the neighbouring forces, predicts 14.99 mm, a margin of 6.2 floors, more than its guard band of 2.0 floors (5.0 under the grid within ±3 floors), so step 6 returns meets. Widened by their guard bands, M2, M5 and M5n return 14.41–15.41, 14.23–15.61 and 14.66–15.33 mm and send the design to trial. The 500 parts had a 95th percentile of 14.88 mm, 8.4 floors below the limit. The case is typical, since at 8.5 floors above the truth step 6 with the nearest setting is right in all 42 interpolated cases, but its guard band was estimated on data that include it.

5.8 Requirements anchored in capability and in tolerances

Upper limits at which an alternative has a performance index of 1.0, 1.33, 1.67 or 2.0 lie a median 1.0, 1.7, 2.5 and 3.1 floors above its 95th percentile; up to an index of 1.33 they lie inside every decisive distance except a near-replicate sibling's (0.85 floors). The test is one-sided by construction: every alternative meets these limits, so a verdict can only be right, a false reject or a trial, and a rule biased towards meets scores well. At an index of 1.33 the shares of right verdicts, false rejects and trials are, with a produced sibling and for a new setting, 91/9/0 and 65/35/0 % for the nearest setting, 44/2/53 and 32/16/52 % for M5, and 81/19/0 and 79/21/0 % for the scaled simulation, which falsely accepts failing alternatives elsewhere; for a new family no rule is right in more than 54 %. Release-level requirements are thus decided at the design stage only with a produced sibling, and then by the production record. General angular tolerances of ISO 2768-1 (ISO, 1989) give a coarser scenario: only 32 of its 108 decisions lie within 5 floors of the limit, 6 of them on failing alternatives, and a cost comparison inherits this coarseness. With a false reject at half the cost of a false accept, the cheapest rule is (i) with a produced sibling, the nearest setting, the scaled simulation and the force trend, tied at zero cost; (ii) for a new setting, the force trend at every trial cost, by three false rejects of the nearest setting among 108 decisions; and (iii) for a new family, a trial of every design while a trial costs at most a tenth of a false accept, the envelope rule at a fifth and M5 beyond.

5.9 Robustness

Table A2 repeats the analysis under variants of the data, constants and model specifications. Adjusting for punch temperature changes no calibrated distance by more than 2.6 floors. Dropping the warm-up of the first 150 parts shrinks the floors and so lengthens the distances, and floors taken between series shorten them. Other floor protocols rescale the decisive distances by 0.60 to 1.42, and the Gaussian-process specifications move them by at most 1.2 floors and the safe distances by at most 2.1. In every variant that includes them, the most decisive rule within a family is the nearest setting or the force trend, and with a produced sibling the nearest setting (tied once). The refitting bootstrap widens the intervals of the learned rules most (M5 4.0–11 floors with a sibling, the nearest setting 2.3–4.6), yet in all 200 resamples (48 configurations) the nearest setting is more decisive than M5 and M5n, with a sibling and for a new setting alike, and the force trend than the scaled simulation (by 1.35–3.6 and 8.8–12 floors); point estimates at the edge of their refit intervals (M1s with a sibling at 4.9 floors, 4.9–7.3, and the safe distances of M2 and M5) reflect what losing a pattern does.

6 Discussion

6.1 What the evaluation establishes

Five findings answer the research questions. Two follow from the definitions and hold wherever the framework is applied: the floor puts every rule and characteristic on a common production-referenced scale, though not on a common scale of nonconforming fractions, and as a within-run quantity it bounds the variation of series production

Table 5 Guidance for design-stage manufacturability decisions, with its basis

Situation	Guidance	Basis
From the framework (prescriptive; usefulness not evaluated)		
Before relying on a predictor	Measure the floor on consecutive production of the family with a stated protocol and report σ_b, σ_w and its lag dependence; treat it as a lower bound of series variation.	framework
Qualifying a rule	Evaluate it for the evidence relation of the coming designs by grouped cross-validation, and split near-replicate siblings from resolvable ones.	framework
Choosing a rule	Compare it with the nearest produced setting and with its own version without the simulation, on decisive and safe distance jointly or by expected cost.	framework
Deciding a single design	Derive the rule's guard band for the distribution of requirement distances the team expects and apply it to the observable margin, in place of range and family guards; a margin inside it means a trial.	framework
Planning a requirement	As a heuristic, plan a trial if the expected margin between requirement and predicted 95th percentile is below the safe distance of the best qualified rule; no decision is reliable closer than a trial reproduces the quantile.	framework
Observed on this dataset (18 alternatives, 2 geometries, 7 characteristics)		
New variant with a produced sibling	nearest produced setting right beyond 3.4 floors (0.85 for near-replicates, 4.9 for resolvable siblings); interval rules safe below one floor, coverage 85–96 %	126 cases
New process setting	nearest setting right beyond 4.7 floors interpolated and 4.9 extrapolated to 500 kN, force trend 8.9 to 100 kN (nearest setting 11; speed change); interval rules safe at 5.9–16 floors, coverage 43–55 %	42 + 84 cases
New family	best rule safe at 16–17 floors in either family's floor, which rule not resolved; rules without the simulation fail; a transferable drawing nominal helps	two transfers
Release-level requirements	lie 1.7 floors above the 95th percentile at an index of 1.33, inside every decisive distance except a near-replicate's; decided only with a produced sibling	126 cases

from below (RQ1); and an interval rule trades decisiveness for safety, which the two distances make explicit, while its distribution-free intervals are guaranteed their level only where the calibration alternatives are exchangeable with the new design. Three are observations on this dataset and hypotheses for confirmation: the best attainable distance grows with the novelty of the design, from 3.4 floors with a produced sibling (the nearest setting 0.85 for near-replicates, 4.9 for resolvable siblings) to 8.2 for a new setting (the force trend; by stratum the nearest setting needed 4.7 interpolated and 4.9 extrapolated to 500 kN, the force trend 8.9 to 100 kN, where the nearest setting needed 11) and 33 for a new family, while within a family the choice of rule matters more than the relation (RQ2); production evidence buys safety before decisiveness, and interval rules held their level with a produced sibling but not when extrapolating or for a new family; and with this simulator and six to nine produced alternatives at two or three force levels, the simulation, as an offset or rescaled, contributed only across families, and M5, M5n and M3n predicted as well as the nearest produced setting with a produced sibling while adding abstention (RQ3).

6.2 Using the framework in design

As a planning heuristic (Table 5), the expected margin between a requirement and the predicted 95th percentile can be compared with the safe distance of the best rule qualified for the relation; if it is smaller, a trial or more production evidence has to be planned, and the decision itself needs the guard band, because the safe distance is defined on the true margin. Release-level requirements make this the usual case, as the practice of a production trial run for a process change expects: the rules decide reliably where a trial is cheapest, a new force on an existing tool being a short press run, and none does near release-level limits where a trial is dearest, for a new family that may need a new tool. The most decisive rule is arguably also the most transparent: the nearest produced setting points to the produced alternatives whose record it averages. The variants that steps 1 to 5 need come from process development and tryout, whose short series of several settings support the sibling and new-setting relations, and from a family's production history across part numbers, which supports the new-family relation; production part approval, whose initial process study covers the released setting, supplies the floor but not the variants.

6.3 Simulation and machine learning in this regime

Within a design family the production record carried the decision, even against a rescaled simulation. The simulated force trend of the two draw-ins was 2.7 to 5.1 times too large, and calibrating the friction coefficient did not repair it, which points to the friction law and the blank-holder model rather than to a parameter. A multi-task model cannot help a family with no produced alternative, because the correlation between families cannot be estimated without its data; in the pooled scopes the Gaussian process with a 0/1 geometry input is a restricted intrinsic coregionalisation model (equal variances, one positive correlation between the families, shared length scales), which improved coverage but not decisiveness. These findings concern one simulator of low fidelity, used through offsets and one scale, and learners with six to nine produced alternatives at two or three force levels. For evaluating AI-based DFM support the lesson is methodological: a learned model should be compared with a production-record baseline, scored by evidence relation in a unit that production defines, and shown to its user with a guard band. A study with design teams could test two hypotheses on reliance (Lee and See, 2004), with cases of known production outcome: designers shown the guard-banded verdict accept a smaller share of wrong verdicts within the safe distance than designers shown the verdict alone, at an equal rate of right acceptances; and, as the same AI assistance helped some teams and hindered others (Zhang et al, 2021), the guard band narrows the spread of that share between teams.

6.4 Implications for engineering design

For variant and platform design (Pahl et al, 2007; Simpson et al, 2006) and the capability databases that serve them (Tata and Thornton, 1999; Thornton, 2004), the results argue for indexing production records by their relation to the coming design rather than for a more sophisticated model on top. The safe distance is the part of a margin that covers predictive uncertainty, as distinct from the part that absorbs change

Table 6 What the framework requires, mapped onto other processes (an untested proposal)

Process	Batch and run	Main variation	Evidence relations	Floor
Sheet forming (this paper)	consecutive parts of one series	drift in a run; coil, die state	variant, setting, family	batch centres in a run
Injection moulding	shots of one cavity and set-up	cavity, shot, set-up, material lot	cavity, process window, new mould	per cavity; cavity offsets as effects
Machining	consecutive parts between offsets	tool wear, offset-compensated	tool path, cutting data, new part	batch centres within an offset interval
Casting	castings of one pattern and melt	melt, pattern position	alloy, gating, new pattern	within a melt; melts as runs
Additive manufacturing	replicates in one build	build, position, orientation	orientation, settings, new part	within a build, via replicated anchors

(Eckert and Isaksson, 2017; Eckert et al, 2019). The growth of the decisive distance, pooled over strata, from about 3 to 8 and over 30 floors says for which overlaps of development (Krishnan et al, 1997; Terwiesch et al, 2002) preliminary manufacturability information can be released, the new-family safe distance, some 0.8 mm of draw-in, how far a set-based design (Sobek et al, 1999) can narrow a feasible set before production evidence exists, and the distances and trial rates when a test is cheaper than a decision (Thomke and Bell, 2001; Salado and Kannan, 2018). The evaluation shows structural validity, that the metrics discriminate between rules and relations on real data (Seepersad et al, 2006); performance validity would require better decisions with the framework.

6.5 Reuse for other processes and AI-based DFM tools

The framework needs several produced variants of a design family whose relations span those of the coming designs, a series of consecutive parts or replicated batches of one design per variant, a design-stage predictor for every variant, and characteristics measured identically on parts and predictions; here rules calibrated on about 50 parts per variant scored within 1.5 floors of those calibrated on full series, against truths and floors from the full series. Attribute outcomes such as splits need a requirement on a rate, which the framework does not yet cover. Other predictors need three adaptations. A classifier with a fixed threshold does not take the requirement as input, so its operating characteristic is estimated by binning test cases by their true margin, not by the exact estimator of Section 3.4, unless the requirement is made an input; a probabilistic classifier needs two thresholds to return a trial. A language-model check may return verdicts that are not monotone in the requirement and that vary between queries, so κ and ω are computed on a requirement grid with repeated queries, and the share of non-monotone verdict sequences is reported. For continuous design spaces the relation becomes a distance from the produced designs in descriptor space or the position relative to their hull, and each stratum needs about as many cases as a relation here: with 126 cases, some 240 verdicts, the 95 % threshold already rests on about a dozen. A generative tool's manufacturability check is scored. A minimal report lists

1. the floor with its protocol, variance components and lag dependence;
2. decisive and safe distances by relation, with refit intervals and one-sided values;

3. coverage and half-width of interval rules, and the source of their noise variance;
4. the nearest-produced-setting and no-simulation baselines;
5. the guard bands with their prior, the trial rates and the performance of the guards;
6. for language-model tools, the prompt and the query protocol.

Table 6, an untested proposal, maps these requirements onto other processes; distances in floors are comparable within a protocol, not across processes, since a floor between builds, cavities or offset intervals is a between-unit quantity (here the floor between series is 1.5 to 9.1 times that within a run). An attempt on open powder bed fusion data of PA12 (Leirmo, 2021) failed on the second requirement: orientation classes mixed orientations in a batch, eight identical anchors per build were too few for a floor within a build, and three builds give three differences between builds.

6.6 Threats to validity

The evaluation covers one material, one press and two geometries, with relations within a family from a 3×3 factorial in which lubrication is mostly not resolved; the procedure is general, the numbers are not. The new family rests on two transfers between geometries that differ in three features at once, and the intervals of its distances hold the two fits fixed. Each alternative was produced once, so the floor covers drift within a run but not between days, coils or tool states, the reference q_{95} is that of one run, and the order and dates of the series are unknown; between-series variation bounds the missing component at 1.2–9.8 floors, and scanner drift cannot be separated from process drift. The extrapolation to 100 kN is confounded with stroke speed, which the simulation omits. The requirements are one-sided limits on the 95th percentile of a characteristic averaged over sides, below the conformance of series release. Several rules, the grouped schemes, the guard bands and their prior were added or chosen after earlier results had been seen and are evaluated on the same data, so the comparison between rules carries selection optimism (Cawley and Talbot, 2010); a single analyst carried out the analysis, with code and data open for replication.

7 Conclusions

Design-stage manufacturability decisions can be evaluated against production on a scale production defines: the decisive and safe distances in floors tell how close to a requirement a rule decides and how close it can be relied on not to mislead, and a guard band derived for the requirements a team expects turns them into a rule for single decisions. On 18 deep-drawn alternatives with matched simulations, the best attainable distance grew with novelty: 3.4 floors with a produced sibling (the nearest setting 0.85 for near-replicates, 4.9 for resolvable ones), 8.2 for a new process setting (the force trend; the nearest setting 4.7 interpolated and 4.9 extrapolated to 500 kN, the force trend 8.9 to 100 kN, with a stroke-speed change), and no rule safe closer than 16–17 floors for a new family, in either family's floor. Within a family the choice of rule mattered more than the relation; requirements at a performance index of 1.33 lay inside every decisive distance except a near-replicate's and need a trial without a produced sibling; and in this small-data regime the simulation as used, offset or

rescaled, contributed only across families, while learned 90 % intervals covered 85–96 % of truths with a produced sibling, under 50 % extrapolating or for a new family.

For design practice, the production record carried the decision within a family, which argues for indexing such records by relation to the coming design rather than for a more elaborate model; across families only rules using the simulation carried the change of geometry, and none decided near release-level limits. For AI-based manufacturability support, the framework offers a standard of evaluation rather than a verdict on machine learning: a tool is scored by evidence relation in floors against a production-record baseline and reported with the checklist of Section 6.5.

The evidence is one process and two geometries, and the comparison of rules is exploratory. Four extensions follow: series from several days, coils and tool states; families with continuous geometric descriptors; a confirmatory test of the frozen procedure on other data; and a study of whether design teams decide better with it.

Appendix A Notation, further results and computation details

Table A1 summarises the notation of Section 3, Table A2 gives the distances under the robustness variants, Table A3 the coverage, width and centre of the interval rules, and Table A4 the procedure's decision rule by relation. Table A5 traces every other number quoted in the text to the script that computes it and the result file that holds it in the archived release (Canseven, 2026), in which `scripts/run_all.sh` runs the scripts in order and `make_tables.py` and `figs.py` write every table and figure from the result files.

Computation details. Batches need at least 80 % valid values. M5 and M5n are Gaussian-process regressions (constant times anisotropic squared-exponential kernel plus white noise, normalised targets, hyperparameters by maximum marginal likelihood without hyperpriors, three restarts, length scales in $[0.1, 100]$) on a geometry indicator, the force in 100 kN and the lubrication rank; the lower bound of the noise variance, the larger of the mean sampling variance of the calibration alternatives' q_{95} and the pooled variance between their series of one geometry and force, is expressed on the normalised scale. NN takes the mean q_{95} at the nearest force of the family, both neighbours when equally near; M1sn fits $q_{95} = a + b\,\text{force}$ by least squares. Mc calibrates, per lubrication pattern, the friction coefficient whose simulated draw-ins are closest to the calibration centres. M3 and M3n use histogram gradient boosting (depth 3, 200 iterations, rate 0.08) on geometry, force, lubrication, sheet thickness and oil film, with 500 incoming conditions drawn from calibration parts of the same pattern; their jackknife+ interval takes the leave-one-out predictions for the new alternative, shifted by the median leave-one-out residual, minus and plus the absolute residuals about that median, with its bounds at the extreme order statistics when n is too small for the 0.9 quantiles. Guard bands lie on a grid of 0.25 floors. The refitting bootstrap draws the three lubrication patterns of each geometry with replacement 200 times, at least two of them distinct, and does not restrict the noise floor to distinct series. Software: Python 3.11 with NumPy, pandas, SciPy, scikit-learn (Pedregosa et al, 2011) and Matplotlib.

Table A1 Notation and terms

Symbol, term	Meaning
a, c, $q_p(a, c)$	design–process alternative, characteristic, p-quantile ($p = 0.95$) of its parts
R, d	requirement (upper limit on c or $\lvert c\rvert$), d floors from the truth: $R = q_p + dF_c$
$[L_r, U_r]$, V_r	interval of rule r for q_p; verdict meets, fails or trial (Eq. 1)
B, m, F_c	batch size (50); batch centre (10 % trimmed mean); production floor (Eq. 2)
σ_b, σ_w	standard deviation between batch means and between parts within a batch
ESR, MRC	effect-to-scatter ratio (Eq. 3); minimum resolvable change, $F_c/\lvert s_c\rvert$
u, v	distances in floors by which an interval lies above or below the truth
$\kappa(\delta)$, $\omega(\delta)$	shares of correct and wrong verdicts among the feasible verdicts at $\lvert d\rvert = \delta$
δ_d, δ_s	decisive distance ($\kappa \geq 0.95$ beyond it) and safe distance ($\omega \leq 0.05$ beyond it)
FA, FR side	distances on the false-accept (failing) and false-reject (adequate) side
g	guard band: observable margin beyond which ≥95 % of decided verdicts were correct, under a stated prior of requirement distances, here the 47 grid distances within ±20 floors, equal weights (Section 3.6); not a property of rule and relation alone
Calibration alternatives	produced alternatives whose record calibrates a rule
Evidence relation	sibling (same setting), new setting (interpolated, extrapolated), new family
Calibration budget	distances against the number k of produced alternatives, by relation
Guard	family, range and novelty checks that flag a design outside the calibration
Measurement resolution	step between adjacent values that the measuring system reports (gauge sense)

Table A2 Decisive/safe distances (floors) under the robustness variants

	New variant				New setting				New family	
Variant	M1s	M2	M5	NN	M2	M5	NN	M1sn	M2	M5
Reference	4.9	23/0.15	7.2/0.75	3.4	18/8.1	19/6.8	8.5	8.2	51/16	38/28
Temperature-adjusted	5.0	23/0.10	7.2/0.65	3.4	17/7.6	19/6.8	9.4	8.9	48/15	37/27
Warm-up dropped	6.5	27/0.20	9.0/0.45	4.1	21/8.7	24/8.7	12	8.8	63/18	39/29
Floor between series	2.9	11/0	1.3/0.20	0.75	9.9/2.6	11/2.3	2.7	4.5	16/3.8	12/9.3
Floor: batches of 25	4.6	21/0.15	6.4/0.70	2.9	16/6.9	17/6.3	7.7	7.3	43/14	31/23
Floor: batches of 100	5.3	24/0.15	8.2/0.80	3.5	19/8.5	21/7.9	8.9	8.9	57/17	46/29
Floor: 90 % quantile	6.2	31/0.20	8.8/0.90	4.0	21/10	23/8.6	10	10	61/20	45/34
Floor: 99 % quantile	3.4	14/0.10	5.2/0.50	2.4	11/4.4	13/4.4	5.7	6.1	37/11	29/15
Calibration: 50 parts	5.0	22/0.20	6.8/0.70	3.4	18/7.5	19/6.2	8.4	7.7	50/16	37/28
Calibration: 100 parts	4.9	22/0.40	6.8/0.40	3.3	18/7.5	19/6.2	8.7	7.8	50/16	38/28
Calibration: 250 parts	4.9	22/0.45	7.0/0.45	3.3	18/7.5	19/6.2	8.7	8.3	50/16	38/28
Conformance $p = 0.90$	4.9	23/0.20	7.2/0.70	3.3	18/7.5	18/6.4	8.4	8.1	51/15	38/28
Conformance $p = 0.99$	5.1	23/0	7.5/0.70	3.3	18/8.3	20/6.9	8.7	8.4	51/16	38/28
Interval level 0.80		23/0.15	6.4/1.0		18/8.1	17/8.1			50/17	38/28
Interval level 0.95		23/0.15	8.0/0.25		18/8.1	20/6.2			51/16	39/27
Response surface	4.8	25/0.30	7.2/0.70	3.4	18/7.6	20/6.9	8.5	8.2	49/25	37/27
GP: sampling bound			6.7/1.4			18/7.4				38/30
Unit: q_{95} floor	5.3	17/0.15	6.9/0.40	2.8	15/6.0	18/6.2	6.9	7.1	41/13	34/30
Refit, δ_d	4.9–7.3	22–23	4.0–11	2.3–4.6	16–18	18–20	6.9–9.2	7.3–10	49–53	34–39
Refit, δ_s	4.9–7.3	0.15–1.4	0.75–3.0	2.3–4.6	6.8–8.4	6.2–7.8	6.9–9.2	7.3–10	15–22	27–33

Rules refitted for every variant; empty cells: the variant does not affect the rule; a new family in its own floor (in the produced family's floor in Table 4). The floor between series contains the lubrication effect and, for a new variant, the series that NN averages, which makes it close to circular there. Warm-up: first 150 parts. Calibration: calibration q_{95} from the first n parts, truths and floors from the full series. Response surface: simulations from a quadratic fit over thickness and friction; other GP kernels and descriptors move M5 by at most 0.4 floors. Unit: the q_{95} floor, 95 % quantile of the difference between the q_{95} of two batches of 100 parts. Refit: 95 % intervals of the decisive and safe distances over 200 draws of each geometry's lubrication patterns with replacement (Section 4.4).

Table A3 Interval rules by relation: coverage, half-width and the interval centre as a point rule

	Sibling produced			Interpolated setting			Extrapolated setting			New family		
Rule	Cov	HW	Ctr	Cov	HW	Ctr	Cov	HW	Ctr	Cov	HW	Ctr
M2	89	7.6	8.6	71	5.7	10	44	4.8	13	47	7.6	34
M2n	90	5.9	9.1	93	5.0	4.8	33	3.1	12	12	5.7	131
M3	96	9.3	9.9	43	7.0	19	43	8.9	21	40	9.7	29
M3n	89	3.0	3.7	36	3.4	11	48	2.3	9.9	7	3.3	130
M5	86	2.2	3.3	71	6.4	9.8	46	4.6	13	12	2.3	34
M5n	85	2.3	3.1	93	5.8	4.7	36	3.7	12	0	2.2	130

Cov: share of cases whose true q_{95} lies in the interval (%; nominal 90). HW: median half-width (floors). Ctr: decisive distance of the interval centre used as a point rule (floors). A new family in the floor of the produced family. The GP noise variance sits at its lower bound in 97 and 99 % of the M5 and M5n fits with a sibling and in every new-setting and new-family fit.

Table A4 The procedure's decision rule (step 6) by relation: guard band, distances and trial shares

	Sibling produced				Interpolated setting				Extrapolated setting			
Rule	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}	g	δ_d/δ_s	T_{10}	T_{20}
M1	none	–	100	100	8.75	20/2.1	44	4	none	–	100	100
M1s	2.0	6.9/2.9	1	0	none	–	100	100	none	–	100	100
M2	0	23/0.15	33	8	0.75	19/2.0	31	4	none	–	100	100
M5	0	7.2/0.75	1	0	0.5	20/2.3	40	4	none	–	100	100
M5n	0	7.2/0.55	0	0	0	16/0	25	0	6.75	25/1.9	48	15
NN	0.25	3.6/3.1	0	0	2.0	6.7/2.7	0	0	7.0	17/2.5	26	1
M3	0	23/0	45	6	17.75	47/0	94	69	16.75	51/0.05	85	69
M3n	0	11/0.05	9	0	6.0	21/1.0	55	6	6.5	19/1.3	47	3
M1sn	0.75	4.3/2.8	0	0	2.0	6.7/2.7	0	0	6.0	15/2.9	21	0

g: guard band (floors) under the reference prior (Section 3.6); no applicability guard is applied; δ_d/δ_s: distances of the rule with its interval widened by g; T_{10}, T_{20}: share of verdicts sent to trial (%) at requirements 10 and 20 floors from the truth. None: no band up to 20 floors reaches 95 %, so the procedure sends every design to trial; so also M0 and M0w in every relation, Mc with a sibling and for an extrapolated setting (for an interpolated one $g = 18$), and every rule for a new family, scored in the produced family's floor.

Table A5 Where the numbers quoted in the text come from: scripts and result files (`results/*.csv`)

Section	Results	Script: result files
3.4, 4.1, 4.3	failing-side weight, run metadata and oil film, scan-line scatter, measurement resolution	panel.py: panel_failing_weight, panel_runs, panel_resolution_steps; measurement.py: meas_scan
5.1	time order, lag profile, normal model, subgroups, part-to-part share, between series, q_{95} reproducibility, in-line drift	alternatives.py: alt_floor; sensitivity.py: sens_drift; panel.py: panel_variogram, panel_floor_protocol; measurement.py: meas_floor; robustness.py: rob_floor_between; inline.py: inline_drift
5.2	resolvable contrasts and changes, simulation offsets, force effect, arm sign, cup depth	alternatives.py: alt_effects, alt_mrc; robustness.py: rob_bhf_pairs; decisions.py: dec_alternatives; panel.py: panel_force_effect, panel_arm_sign, panel_m0_split
5.3–5.4	distances in both floors, paired differences, strata, force levels, decomposition, pooled coverage	decisions.py: dec_resolution, dec_paired, dec_coverage; panel.py: panel_source_floor(_paired), panel_sibling_strata, panel_force_levels, panel_decomposition
5.5	simulation pairs, drawing nominal, surrogate, coverage, GP fits	decisions.py: dec_paired, dec_coverage, dec_gp; panel.py: panel_nominal_offset, panel_source_floor(_paired); design_space.py: ds_surrogate(_rule)
5.6, 5.9	budget, short series, robustness variants, refit, q_{95} unit	budget.py: budget_design, budget_resolution; robustness.py: rob_decisions, rob_refit(_paired); panel.py: panel_q95_unit
3.6, 5.7	margins, guard bands and priors, uncapped bands, guards, worked example	panel.py: panel_step6(_curves), panel_guard_priors; guard.py: dec_guard; decisions.py: dec_intervals
5.8	capability-anchored limits, ISO 2768-1 scenario, costs	panel.py: panel_capability; scenarios.py: scen_strata, scen_scores, scen_cost_map
6.5	PA12 attempt	case2_pa12.py: case2_floor, case2_effects

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Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. The extracted per-part and per-simulation feature tables and all result files are archived with the code (Canseven, 2026).

Code availability. The complete analysis pipeline, which reproduces every number in this paper from the public data, is archived with the feature tables and result files of this version on Zenodo (Canseven, 2026); Table A5 maps the numbers to its scripts and files. A working copy is at https://github.com/htcanseven/Claude (directory `scripts/`).

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) assisted under the author's direction, as Section 4.5 describes. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. All figures are produced by the released code; no AI-generated images are used.

References

Angelopoulos AN, Bates S, Fisch A, et al (2024) Conformal risk control. In: International Conference on Learning Representations (ICLR 2024). OpenReview.net, URL https://openreview.net/forum?id=33XGfHLtZg

ASME (2018) Assessing credibility of computational modeling through verification and validation: Application to medical devices. ASME V&V 40-2018, American Society of Mechanical Engineers, New York

Automotive Industry Action Group (2006) Production Part Approval Process (PPAP), 4th edn. AIAG, Southfield, MI

Automotive Industry Action Group (2010) Measurement Systems Analysis, 4th edn. AIAG, Southfield, MI

Ballegeer M, Benoit DF (2026) BenDFM: A taxonomy and synthetic CAD dataset for manufacturability assessment in sheet metal bending. Journal of Intelligent Manufacturing https://doi.org/10.1007/s10845-026-02857-9

Banabic D (2010) Sheet Metal Forming Processes: Constitutive Modelling and Numerical Simulation. Springer, Berlin, Heidelberg, https://doi.org/10.1007/978-3-540-88113-1

Barber RF, Candès EJ, Ramdas A, et al (2021) Predictive inference with the jackknife+. The Annals of Statistics 49(1):486–507. https://doi.org/10.1214/20-aos1965

Baum S, Heinzelmann P (2025) Deep drawing and cutting simulations (DDACS) dataset. DaRUS, University of Stuttgart, version 3.0, https://doi.org/10.18419/DARUS-4801

Baum S, Heinzelmann P (2026) Real deep drawing and cutting (RDDAC) dataset. DaRUS, University of Stuttgart, version 2.0, https://doi.org/10.18419/DARUS-5589

Baum S, Heinzelmann P, Clauß P, et al (2026a) Statistical analysis of simulation-to-reality deviation in deep drawing with a benchmark dataset. Transactions of the Indian Institute of Metals 79(7):176. https://doi.org/10.1007/s12666-026-03870-5

Baum S, Heinzelmann P, Liewald M, et al (2026b) Addressing discrepancies between synthetic and real-world data of deep-drawing manufacturing systems in AI applications. Procedia CIRP 138:727–732. https://doi.org/10.1016/j.procir.2026.01.125

Bechhofer RE (1954) A single-sample multiple decision procedure for ranking means of normal populations with known variances. The Annals of Mathematical Statistics 25(1):16–39. https://doi.org/10.1214/aoms/1177728845

Boothroyd G, Dewhurst P, Knight WA (2010) Product Design for Manufacture and Assembly, 3rd edn. CRC Press, Boca Raton, https://doi.org/10.1201/9781420089288

Canseven HT (2026) Code, feature tables and results for "evaluating AI-based manufacturability decisions against the resolution of production". Zenodo, version 1, *[DOI pending]*

Cawley GC, Talbot NLC (2010) On over-fitting in model selection and subsequent selection bias in performance evaluation. Journal of Machine Learning Research 11:2079–2107. URL https://jmlr.org/papers/v11/cawley10a.html

Chow CK (1970) On optimum recognition error and reject tradeoff. IEEE Transactions on Information Theory 16(1):41–46. https://doi.org/10.1109/TIT.1970.1054406

Dreyfus PA, Psarommatis F, May G, et al (2022) Virtual metrology as an approach for product quality estimation in Industry 4.0: a systematic review and integrative conceptual framework. International Journal of Production Research 60(2):742–765. https://doi.org/10.1080/00207543.2021.1976433

Eckert C, Isaksson O (2017) Safety margins and design margins: A differentiation between interconnected concepts. Procedia CIRP 60:267–272. https://doi.org/10.1016/j.procir.2017.03.140

Eckert C, Isaksson O, Earl C (2019) Design margins: a hidden issue in industry. Design Science 5:e9. https://doi.org/10.1017/dsj.2019.7

Friedman JH (2001) Greedy function approximation: A gradient boosting machine. The Annals of Statistics 29(5):1189–1232. https://doi.org/10.1214/aos/1013203451

Geifman Y, El-Yaniv R (2017) Selective classification for deep neural networks. In: Advances in Neural Information Processing Systems, vol 30. Curran Associates, Inc., pp 4878–4887

Ghadai S, Balu A, Sarkar S, et al (2018) Learning localized features in 3D CAD models for manufacturability analysis of drilled holes. Computer Aided Geometric Design 62:263–275. https://doi.org/10.1016/j.cagd.2018.03.024

Hazelrigg GA (1998) A framework for decision-based engineering design. Journal of Mechanical Design 120(4):653–658. https://doi.org/10.1115/1.2829328

Heddoub A, Diallo AR, Homri L, et al (2026) Conformal prediction for multivariate quality assurance – application on coordinate measuring machines. Procedia CIRP 145:91–96. https://doi.org/10.1016/j.procir.2026.03.151

Heinzelmann P, Baum S, Riedmüller KR, et al (2025) A comprehensive benchmark dataset for sheet metal forming: Advancing machine learning and surrogate modelling in process simulations. MATEC Web of Conferences 408:01090. https://doi.org/10.1051/matecconf/202540801090

Hol J, Cid Alfaro MV, de Rooij MB, et al (2012) Advanced friction modeling for sheet metal forming. Wear 286–287:66–78. https://doi.org/10.1016/j.wear.2011.04.004

ISO (1989) General tolerances – part 1: Tolerances for linear and angular dimensions without individual tolerance indications. ISO 2768-1:1989, International Organization for Standardization, Geneva

ISO (2017) Geometrical product specifications (GPS) – inspection by measurement of workpieces and measuring equipment – part 1: Decision rules for verifying conformity or nonconformity with specifications. ISO 14253-1:2017, International Organization for Standardization, Geneva

ISO (2023) Accuracy (trueness and precision) of measurement methods and results – part 3: Intermediate precision and alternative designs for collaborative studies. ISO 5725-3:2023, International Organization for Standardization, Geneva

ISO (2025) Accuracy (trueness and precision) of measurement methods and results – part 2: Basic method for the determination of repeatability and reproducibility of a standard measurement method. ISO 5725-2:2025, International Organization for Standardization, Geneva

ISO (2026) Statistical methods in process management – capability and performance – part 2: Process capability and performance of time-dependent process models. ISO 22514-2:2026, International Organization for Standardization, Geneva

Kennedy MC, O'Hagan A (2000) Predicting the output from a complex computer code when fast approximations are available. Biometrika 87(1):1–13. https://doi.org/10.1093/biomet/87.1.1

Kennedy MC, O'Hagan A (2001) Bayesian calibration of computer models. Journal of the Royal Statistical Society Series B: Statistical Methodology 63(3):425–464. https://doi.org/10.1111/1467-9868.00294

Krishnan V, Eppinger SD, Whitney DE (1997) A model-based framework to overlap product development activities. Management Science 43(4):437–451. https://doi.org/10.1287/mnsc.43.4.437

La Trobe-Bateman J, Wild D (2003) Design for manufacturing: use of a spreadsheet model of manufacturability to optimize product design and development. Research in Engineering Design 14(2):107–117. https://doi.org/10.1007/s00163-003-0030-8

Lee JD, See KA (2004) Trust in automation: Designing for appropriate reliance. Human Factors: The Journal of the Human Factors and Ergonomics Society 46(1):50–80. https://doi.org/10.1518/hfes.46.1.50.30392

Lei J, G'Sell M, Rinaldo A, et al (2018) Distribution-free predictive inference for regression. Journal of the American Statistical Association 113(523):1094–1111. https://doi.org/10.1080/01621459.2017.1307116

Leirmo TL (2021) Data on dimensional and geometric accuracy for laser-based powder bed fusion of PA12 (PA2200). DataverseNO, version 1.1, https://doi.org/10.18710/DHACHZ

Madras D, Pitassi T, Zemel R (2018) Predict responsibly: Improving fairness and accuracy by learning to defer. In: Advances in Neural Information Processing Systems, vol 31. Curran Associates, Inc., pp 6147–6157

Makatura L, Foshey M, Wang B, et al (2024) How can large language models help humans in design and manufacturing? Part 1: Elements of the LLM-enabled computational design and manufacturing pipeline. Harvard Data Science Review (Special Issue 5). https://doi.org/10.1162/99608f92.cc80fe30

Malak RJ, Paredis CJJ (2010) Using support vector machines to formalize the valid input domain of predictive models in systems design problems. Journal of Mechanical Design 132(10):101001. https://doi.org/10.1115/1.4002151

Montgomery DC (2019) Introduction to Statistical Quality Control, 8th edn. Wiley, Hoboken, NJ

Nemani V, Biggio L, Huan X, et al (2023) Uncertainty quantification in machine learning for engineering design and health prognostics: A tutorial. Mechanical Systems and Signal Processing 205:110796. https://doi.org/10.1016/j.ymssp.2023.110796

Oberkampf WL, Roy CJ (2010) Verification and Validation in Scientific Computing. Cambridge University Press, Cambridge, https://doi.org/10.1017/cbo9780511760396

Pahl G, Beitz W, Feldhusen J, et al (2007) Engineering Design: A Systematic Approach, 3rd edn. Springer, London, https://doi.org/10.1007/978-1-84628-319-2

Pedregosa F, Varoquaux G, Gramfort A, et al (2011) Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12(85):2825–2830. URL https://jmlr.org/papers/v12/pedregosa11a.html

Picard C, Edwards KM, Doris AC, et al (2025) From concept to manufacturing: evaluating vision-language models for engineering design. Artificial Intelligence Review 58(9):288. https://doi.org/10.1007/s10462-025-11290-y

Rasmussen CE, Williams CKI (2006) Gaussian Processes for Machine Learning. MIT Press, Cambridge, MA, https://doi.org/10.7551/mitpress/3206.001.0001

Regenwetter L, Nobari AH, Ahmed F (2022) Deep generative models in engineering design: A review. Journal of Mechanical Design 144(7):071704. https://doi.org/10.1115/1.4053859

Roberts DR, Bahn V, Ciuti S, et al (2017) Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. Ecography 40(8):913–929. https://doi.org/10.1111/ecog.02881

Salado A, Kannan H (2018) A mathematical model of verification strategies. Systems Engineering 21(6):593–608. https://doi.org/10.1002/sys.21463

Seepersad CC, Pedersen K, Emblemsvåg J, et al (2006) The validation square: How does one verify and validate a design method? In: Lewis KE, Chen W, Schmidt LC (eds) Decision Making in

Engineering Design. ASME Press, New York, p 303–313, https://doi.org/10.1115/1.802469.ch25
Simpson TW, Siddique Z, Jiao JR (eds) (2006) Product Platform and Product Family Design: Methods and Applications. Springer, New York, https://doi.org/10.1007/0-387-29197-0
Sobek DK, Ward AC, Liker JK (1999) Toyota's principles of set-based concurrent engineering. Sloan Management Review 40(2):67–83. URL https://sloanreview.mit.edu/article/toyotas-principles-of-setbased-concurrent-engineering/
Tahera K, Wynn DC, Earl C, et al (2019) Testing in the incremental design and development of complex products. Research in Engineering Design 30(2):291–316. https://doi.org/10.1007/s00163-018-0295-6
Tata M, Thornton AC (1999) Process capability database usage in industry: Myth vs. reality. In: Proceedings of the ASME 1999 Design Engineering Technical Conferences, Volume 4: 4th Design for Manufacturing Conference. American Society of Mechanical Engineers, pp 395–405, https://doi.org/10.1115/detc99/dfm-8968
Tercan H, Meisen T (2022) Machine learning and deep learning based predictive quality in manufacturing: a systematic review. Journal of Intelligent Manufacturing 33(7):1879–1905. https://doi.org/10.1007/s10845-022-01963-8
Terwiesch C, Loch CH, De Meyer A (2002) Exchanging preliminary information in concurrent engineering: Alternative coordination strategies. Organization Science 13(4):402–419. https://doi.org/10.1287/orsc.13.4.402.2948
Thomke S, Bell DE (2001) Sequential testing in product development. Management Science 47(2):308–323. https://doi.org/10.1287/mnsc.47.2.308.9838
Thomke SH (1998) Simulation, learning and R&D performance: Evidence from automotive development. Research Policy 27(1):55–74. https://doi.org/10.1016/s0048-7333(98)00024-9
Thornton AC (2004) Variation Risk Management: Focusing Quality Improvements in Product Development and Production. Wiley, Hoboken, NJ
Thornton AC, Donnelly S, Ertan B (2000) More than just robust design: Why product development organizations still contend with variation and its impact on quality. Research in Engineering Design 12(3):127–143. https://doi.org/10.1007/s001630050028
Tibshirani RJ, Barber RF, Candès EJ, et al (2019) Conformal prediction under covariate shift. In: Advances in Neural Information Processing Systems, vol 32. Curran Associates, Inc., pp 2530–2540
Whitney DE (1996) Why mechanical design cannot be like VLSI design. Research in Engineering Design 8(3):125–138. https://doi.org/10.1007/bf01608348
Wuest T, Weimer D, Irgens C, et al (2016) Machine learning in manufacturing: advantages, challenges, and applications. Production & Manufacturing Research 4(1):23–45. https://doi.org/10.1080/21693277.2016.1192517
Xie J, Safdar M, Mircea A, et al (2025) Towards reproducible machine learning-based process monitoring and quality prediction research for additive manufacturing. Engineering Applications of Artificial Intelligence 161:112223. https://doi.org/10.1016/j.engappai.2025.112223
Xie J, Yang Z, Yang HC, et al (2026) On the reusability of machine learning-based process monitoring systems for manufacturing digital twins. Journal of Manufacturing Systems 84:333–356. https://doi.org/10.1016/j.jmsy.2025.12.006
Yan X, Wang Z, Zhao C, et al (2026) Machine learning-based cyber manufacturing services: A review of manufacturing process selection, process planning, and design for manufacturing. Engineering 63:47–67. https://doi.org/10.1016/j.eng.2025.11.034
Yoshida F, Uemori T (2002) A model of large-strain cyclic plasticity describing the Bauschinger effect and workhardening stagnation. International Journal of Plasticity 18(5–6):661–686. https://doi.org/10.1016/S0749-6419(01)00050-X
Zhang G, Raina A, Cagan J, et al (2021) A cautionary tale about the impact of AI on human design teams. Design Studies 72:100990. https://doi.org/10.1016/j.destud.2021.100990
Zhang Z, Jaiswal P, Rai R (2018) FeatureNet: Machining feature recognition based on 3D convolution neural network. Computer-Aided Design 101:12–22. https://doi.org/10.1016/j.cad.2018.03.006
Zong Y, Badakhshan S, Zhang J, et al (2026) Trustworthy uncertainty quantification via distributionally-robust dual adaptive conformal prediction. Journal of Mechanical Design 148(2):021703. https://doi.org/10.1115/1.4069683